

POLARIS DC-1 — TECHNICAL SPECIFICATION

SPEC-02: STRUCTURAL STEEL AND PRECAST CONCRETE

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, values and figures in this document are fictional.

1. Scope

- 1.1 This specification covers the fabrication, supply and erection of the primary and secondary structural steel frame, and the supply and erection of precast concrete wall panels forming the external envelope inner leaf and the inter-hall fire-separating walls, for all buildings on the 12 m × 18 m primary grid defined in PDC-BOD-001 §2.1.
- 1.2 The frame is a steel braced-portal frame, clear-span across the 18 m dimension within each data hall (no internal columns within white space), on 12 m column centres. The shared MEP spine and the two-storey Administration/NOC block use the same grid with an intermediate column line.
- 1.3 This specification excludes envelope insulation, rainscreen cladding and roofing membranes (PDC-SPEC-03) and internal finishes and containment (PDC-SPEC-04), which are fixed to, or supported by, the structure specified here.

2. Standards and References

Reference	Subject
SFS-EN 1993-1-1, -1-3, -1-8 (Eurocode 3), Finnish National Annexes	Design of steel structures
SFS-EN 1990, SFS-EN 1991-1-1/-1-3/-1-4, Finnish National Annexes	Basis of design, snow and wind, Vantaa location
SFS-EN 1090-1, SFS-EN 1090-2	Execution of steel structures — CE marking and technical execution requirements
SFS-EN 10025 (parts 1–6)	Structural steel grades and technical delivery conditions
SFS-EN ISO 3834-2	Welding — quality requirements for fusion welding
SFS-EN 1992-1-1 (Eurocode 2), Finnish National Annex	Design of concrete structures
SFS-EN 13369	Common rules for precast concrete products
SFS-EN 14992	Precast concrete — wall elements
SFS-EN 13501-2	Fire classification of construction products — resistance to fire (EI ratings)

National Building Code of Finland	Fire compartmentation and structural fire safety
SFS-EN 12812	Falsework — performance requirements (temporary works during erection)

Where SFS-EN 1090-2 requires an Execution Class (EXC) to be specified: EXC3 applies to primary frame members, connections and bracing in all buildings; EXC2 applies to secondary steel, purlins and rails.

3. Materials

Element	Specification
Primary columns	Universal Column (UC) sections, steel grade S355 J2
Primary rafters/trusses (18 m span)	Fabricated lattice trusses, steel grade S355 J2, depth coordinated to keep underside of steel at 6.6 m above finished floor level
Secondary steel (purlins, rails, bracing)	Steel grade S355 or S275 as designed, cold-formed or hot-rolled
Bolted connections	Grade 8.8 general, grade 10.9 at primary moment/bracing connections, per SFS-EN 14399
Fire protection to structural steel	Intumescent coating or board, rated to the fire-resistance period required by the fire-engineering submission for each area
Precast wall panels	Reinforced concrete, minimum C40/50, to SFS-EN 14992
Panel joints (all locations)	Gasketed and sealed at 3.0 m centres — this refines the BOD-stage 2.4 m panel-width estimate (PDC-BOD-001 §4.4) and is the figure the precast subcontractor shall detail at shop-drawing stage
Inter-hall separating wall panels	As above, achieving not less than EI 120 , continuous from foundation to underside of roof deck
Panel fixings and connections	Stainless steel or hot-dip galvanized, designed for differential movement between panel and frame
Fire-stopping at penetrations (busway, pipework, cable routes crossing a compartment line)	Rated to match the EI 120 wall it penetrates, third-party tested system

4. Workmanship and Execution

4.1 Fabrication. All primary steelwork shall be fabricated by a workshop holding SFS-EN 1090-1 CE marking certification for the relevant Execution Class. Welding procedures shall be qualified to SFS-EN ISO 15614 and welders qualified to SFS-EN ISO 9606.

4.2 Erection sequence. Erection shall follow the Section sequence set out in PDC-BOD-001 §2.3 (west to east), with temporary hoarding maintained on the numbered gridline immediately east of the most recently completed hall. The Contractor shall not erect steelwork for a later Section in a manner that requires access through, or imposes loads on, an operating Section's structure without a live-environment method statement.

4.3 Bracing. Horizontal and vertical bracing bays shall be located at the gridline intersections adjacent to each fire-compartment wall, so that no bracing bay crosses a compartment line, per PDC-BOD-001 §4.1.

4.4 Precast panel erection. Panels shall be erected plumb and true to line, with joints gasketed and sealed progressively as erection proceeds — panels shall not be left with open joints exposed to weather beyond 5 working days. The four inter-hall separating walls shall be erected as a continuous run from foundation to roof deck before the adjacent halls' fit-out

trades commence in either hall.

4.5 Fire-stopping. All penetrations through an EI 120 wall shall be fire-stopped using a system tested in that wall configuration; fire-stopping shall not be covered or concealed by following trades until inspected and recorded (Section 6).

4.6 Coordination with envelope. The rainscreen support system and roof deck (PDC-SPEC-03) shall be coordinated with the primary and secondary steel at shop-drawing stage so that no additional penetrations are required in the EI 120 walls after erection.

5. Tolerances

Item	Tolerance (per SFS-EN 1090-2 unless noted)
Column position in plan	± 10 mm
Column verticality (per storey height)	1:500, maximum 10 mm
Column level (base plate)	± 5 mm
Rafter/truss level (bearing)	± 10 mm
Overall frame verticality (full height)	1:1000
Precast panel plan position	± 10 mm
Precast panel plumb (per panel height)	1:400
Precast panel joint width	15 mm ± 5 mm (nominal 3.0 m centres)
Precast panel face alignment between adjacent panels	± 5 mm

6. Testing and Inspection

Test / inspection	Frequency	Acceptance basis
Weld non-destructive testing (visual, MT/UT as designed)	Per SFS-EN ISO 17635 sampling plan, 100% of primary connections classed critical	No unacceptable indications per SFS-EN ISO 5817, quality level B for primary members
Bolt torque/tension inspection	100% of primary moment and bracing connections	Torque or tension within specified range, per SFS-EN 1090-2
Dimensional survey of erected frame	On completion of each Section's frame	Within tolerances in Section 5
Fire protection coating thickness (DFT)	10% of coated members, minimum 3 per zone	Meets the certified fire-test data for the specified rating
Fire-stopping inspection and certification	100% of penetrations through EI 120 walls	Installed system matches the certified test configuration; photographic record retained
Precast panel joint/EI 120 verification	Sample inspection, minimum one location per inter-hall wall	Consistent with the third-party fire test report for the panel/joint system

7. Submittals

Submission	Timing
Shop and erection drawings, connection design calculations	S3 construction issue
Welding Procedure Specifications (WPS) and welder qualification records	Before fabrication commences
Mill certificates for structural steel	With each delivery batch
Erection method statement, including sequence and temporary works (if any)	S3 construction issue

Erection method statement, including sequence and temporary works (falsework, propping)	S3, updated per Section
Fire-engineering submission for EI 120 wall/joint/penetration details	S2 detailed design, before precast fabrication
Precast panel design calculations and shop drawings	S3 construction issue
Fire-stopping product data and third-party test certificates	Before installation
As-built survey and O&M documentation	Per Section handover

8. Warranties

8.1 The structural frame and precast envelope shall be designed for a 50-year design life for the primary structure and a 25-year design life for the envelope, per PDC-BOD-001 §4.3.

8.2 Fire protection coatings and fire-stopping systems shall carry the manufacturer's standard warranty, and the Contractor shall retain spare touch-up paint (minimum 5 litres per colour) and spare fire-stopping material sufficient for future service penetrations, handed over with each Section's O&M documentation.

End of PDC-SPEC-02 Rev A.

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